CHAPTER 1

An educational dungeon crawler designed to encourage young people to write their own stories and support their creative writing process.

Game Design Document

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1. Game Overview

1.1 Game Concept

Chapter 1 is an educational dungeon crawler, in which players guide a self-designed adventurer through different worlds, encountering enemies, puzzles and other obstacles. Rather than using combat, items or other RPG staples to pass obstacles, players must write out their own solutions using nothing more than a keyboard and their imagination.

The game uses natural language processing to generate new writing prompts based on what the player has already written. Everything the player writes is saved into one document, so that as players progress through each dungeon they build up a story. This can be read and downloaded at the end of the game.

1.2 Target Audience

Chapter 1 is targeted at the education market; it should be a game that teachers want to utilise in the classroom to support students with their creative writing and literacy. As a result the game will need to:

a) be appealing to children and young people between the ages of 8-13

b) demonstrate pedagogical value, both as a standalone exercise and part of wider structured learning.

1.3 Design Values

The following values will guide the design of the game:

- **Approachable**

First and foremost I want the game to be attractive to young people; despite being an educational game, this should be something that they want to play. I imagine a lot of this will be achieved through aesthetics – cartoonish visuals, satisfying effects, and an exciting world to explore. The core game loop should also be fun to play, and encourage repeat playthroughs.

- **Accessible**

I want to create a game that is easy for young people to play, even if they might not be the top of their class in Literacy. The whole point of learning games is just that – they are about learning. The game should not be limited to those students who have learned a lot already.

- **Structured creativity**

Research shows that young people respond well to being given the freedom to express themselves. However, complete freedom means that learning objectives are not met. This is why the concept of 'structured creativity' is so important to this project – I want to give players freedom to express

themselves through writing, but build a structure to this expression so that they have a tangible end product at the end of the game.

- **Encouraging and inspiring**

I want the game to inspire players to create their own stories outside of the game, after playing. After completing the game players will be presented with their complete story, and will have the option to print this out. They will be encouraged to share this story with others, and also given tips on how to improve. Positive reinforcement is really important when encouraging young people to persevere with something they might find a challenge.

- **Impactful learning experience**

As an education game first and foremost, this project should provide an impactful learning experience for players. Alongside giving players the kind of scaffolding to start and continue writing their story (through the generated prompts), the game should offer guidance on how players can improve their writing.

- **Encourage adult involvement**

The problem with a lot of learning games is that they are seen as an activity for young people to do on their own – a way of learning without any adult input. I don't believe this is an effective way of supporting a young person to learn. I want this game to encourage adults to get involved in the learning process, supporting the young person as they write their stories, and then reading the finish product afterwards as well.

1.4 Features

The following core features will be central to the game:

- **Character customisation**. Players should be able to customise their character prior to beginning the game.
- **Overworld exploration.** Players will explore a large game world with branching paths, so they have control over their own adventure. The world will contain different obstacles that block each path, forcing the player to engage with…
- **Interactable objects**. Upon interacting with objects in the game world the play will be prompted to write about how they get past the object. Only when a number of written prompts have been completed with the player be able to proceed.
- **Natural language processing prompts.** Writing prompts are generated using natural language processing create prompts that adapt to what the player has already written. This ensures that no playthrough is the same, and that the player receives more tailored scaffolding for their story.
- **Support prompts**. After answering the initial prompts the player may receive addition ideas e.g. 'Try using a new adjective' to support their writing. Not yet clear how prevalent this feature will be.

- **Multiple exits**. The game will contain multiple exits that the play can find by navigating the different paths. These exits will have an impact on the ending of the player's story.
- **Collated story.** Everything that the player writes will be collated into a complete story that they can then read and download at the end of the play session.

The game will also have the following additional features:

- **Unlockable character options.**
- **Unlockable cosmetics.**
- **Multiple game worlds.**
- **Progress tracking.**

1.5 Visual References

I want the game to be viable on school laptops rather than high-end machines, so the graphical style won't be too complex. The game will take place in a 4:3 ratio window.

The overworld will be made out of isometric tiles, with a low-detail pixelated look, similar to that used in *Into The Breach* (right).

The overworld will be occupied by different obstacles, all given a tile. They might be built into the tile, such as a crevasse or a mountain, or they might just be an object or NPC on-top of the tile. These will be pixel sprites, styled in 3D to fit the isometric POV.

Rather use 3D sprites, I will make the player character a 2D sprite so that they stand out from the backdrop. I think this will also make the movement feel a lot freer and less stuck to the tiles. The movement of old Lucas Arts adventure games is a good reference point (*Ben There Dan That* (below left) captures this depth within a 3D environment using the Adventure Game Studio toolset. More recently, *Broken Age* (below right) did a great job of using flat art but giving the world depth.

The game will include illustrated portraits of characters, including the player character. These will be used in dialogue sections.

Before beginning the game, players will be able to design their own characters in a Mii-like character designer. There will be a number of options for hair, eyes, mouth, etc, with the potential for about 12 different faces (this can easily be expanded on beyond the initial prototype). These portraits will have a hand-painted cartoon look, similar to *Animal Crossing* villagers (right).

2. Research

I did a great deal of research into creativity and how we can encourage it, especially through games, before beginning work on this project.

2.1 Workshops

As part of my early design process I ran four half-hour design workshops with small groups of primary school children aged 8-11.

The workshops were all half an hour in length so as to maximise students' attention for that time. Workshops were run on an opt-in basis; students could choose whether they wanted to attend or not. This ensured that all students attending were inherently motivated to complete the tasks, and limited the need for behaviour management during the sessions. Students had no prior knowledge of the content of the workshop (it was billed as a 'fun creative writing workshop' on parent letters and in conversations with students).

Each workshop examined different methods of teaching creative writing, primarily game mechanics, and students' responses to these methods. Below I shall provide details of each workshop, including the outcomes I took away from the session.

Design Workshop Questionnaire Responses

Before each workshop I asked attendees to fill out a quick questionnaire which asked the following questions:

1. On a scale of 1-5, how much do you enjoy reading?
2. On a scale of 1-5, how much do you enjoy creative writing?
3. What do you like/dislike about creative writing?
4. How you make creative writing more fun?
5. If you could be anything in the world, what you could be, and why?

The first two questions gave me a sense of the student's engagement before the session began. This was used to contextualise their enthusiasm for the session and the different game mechanics. If a student states they do not enjoy creative writing, but are enthusiastic about a certain mechanic, then I'll know that mechanic is of particular relevance to the experience I'm looking to create and the learning objectives I'm looking to meet with **Chapter 1**.

The third question was also relevant to this aim – I did not want to include anything in my game that would put students off creative writing.

The fourth question got students to start thinking about how they might make creative writing tasks more fun. What was interesting is that this was the question that students struggled with most; they found it difficult to imagine a creative writing activity outside of those already undertaken at school. The idea that creative writing could be 'fun' was alien to them, validating the need for a project like **Chapter 1**.

The fifth question was designed as a warm-up for students before they began the creative writing games that the session contained.

Workshop #1

The first workshop was focused on how structured writing prompts can support or inhibit creative writing. It was deliberately quite a staid session, utilising 'traditional' educational methods to serve as a control for later workshops.

First, students were asked to complete a writing task with a pre-created prompt based on Key Stage 2 standardised writing tests. Additional prompts were provided alongside this to guide students on what needed to be included in the story.

Students were then were asked to complete a free-writing task, in which they were given a blank piece of paper and told to write a story. They were given no further prompts, and 5 minutes to complete their story.

From these two activities I learnt that giving students the ability to choose what they write about generated much more enthusiasm for the task, and eventual creativity. However younger, less confident students struggled to get started and get past their 'blank page anxiety'. This suggests there is a value to prompts that serve as scaffolding without strangling the potential for creativity. The situation also showcases the importance of adaptive difficulty within learning games, to ensure that tasks are accessible to all.

Workshop #2

The second workshop combined free-writing with the first game mechanic, a randomised story prompt generator, to compare the two. This session was difficult to manage due to the number of students who weren't keen on creative writing.

The second task utilised a set of dice with different icons on. Students took it in turns to roll the dice, and then had one minute to write a part of a story that involved the icon on the dice. When the time is up, the next student rolls the dice, and students have to continue the story, now involving the new icon. This task was designed to encourage students to think creatively rapidly – it was a test of a random prompt generation idea, and also a time limit.

The random prompt generation lead to some outlandish and entertaining stories, however students complained that they didn't have enough time to complete their stories. Students were also often distracted by rolling the dice (the game mechanic), often to the detriment of their story. It became apparent that I would need to ensure that none of the gameplay mechanics obscured the learning objectives in **Chapter 1**.

Workshop #3

The third workshop was focused on character creation, and looked into whether students with a greater connection to their character showed a strong desire to create a compelling storyline for that character. I also used this workshop to investigate the potential of visual storytelling.

Students were first asked to create a character through a game of exquisite corpse, in which players draw a head, fold the paper to hide their drawing, and then pass the paper on for the next player to continue the body, and so on. Each student was left with a completed character. They were then asked to write a story about this character in five minutes.

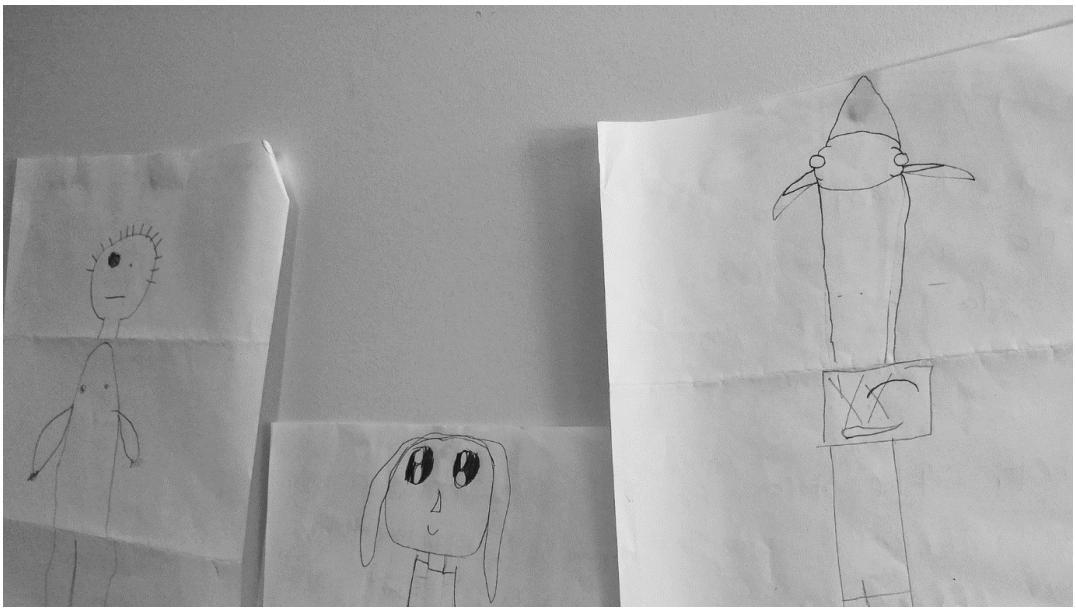


Figure 1 - Exquisite Corpse Examples

The second activity saw students given a short 'character sheet' which asked students questions about their character such as Name, Age, Profession, etc. A longer field, Description, was also present. Students were asked to complete this sheet in five minutes, creating a character from scratch.

Participants were then asked to write a story about their newly designed characters. Even though there were no visuals of the character, students were much more amenable to this task than the first. I attribute this to the fact that students had control over the character they were going to write about. They felt a greater ownership over their story; for young people, the concept of the 'main character' in a story is an important one, for reasons detailed above – young people like to project themselves into the story. This is why games are such a powerful medium for engaging them.

Workshop #4

The final workshop overlapped with the early development of my first paper prototype, so I ended up testing this in this workshop, alongside a more traditional creative writing task to compare the outcome.

Students were first asked to complete a written task very similar to in the first workshop. They were given a written prompt taken from a standardised creative writing test, and asked to write a story that fulfilled the prompt. This prompt was slightly more open to interpretation: it asked students to imagine a new animal, and create a guide for looking after that animal.

For the second activity I introduced the paper prototype I had created. The game was a simple board game, with players rolling a dice to progress around the board. However, each square was denoted a different type of prompt: Setting, Character, Obstacle or Treasure. Every time a player landed on a

square they had to pick-up a prompt and write it into their story. Each player started with a setting card to get them started, and they were able to write about a character of their choosing, either fictional or real.

This was the workshop during which I realised I had the basics of a strong gameplay loop. There was enough freedom for students to write what enthused them, as well as enough structure to ensure they didn't feel lost. They ended up producing a coherent story as the board progressed in a manner that introduced end states as they reached the end (Treasure).

2.2 Prototyping

My next task was to iterate on the paper prototype developed for my design workshops and test it further.

I introduced a couple of new rules to the version I ran with the students. Players were now no longer able to change their setting midway through the game, as I found this derailed the story. I also introduced some new scaffolding for creating characters in the form of a character sheet, as some players were struggling with this aspect of the game.

Figure 2 – Board Game Prototype

I tested this prototype further with two groups of older testers (20s-30s rather than primary school children). What affirmed my idea was how they also responded positively to the structure of the game, and worked within it to exercise their creativity and produce stories of their own.

Two pieces of feedback came through from these sessions. Firstly, that traversing the board became boring after a while and didn't feel like 'exploring'. Secondly, the prompts were a bit too open-ended, and could be tightened up to further help players who weren't experienced in writing or telling stories.

I took the feedback from these playtest sessions and iterated on them in my first plan for the digital version of the game.

3. Gameplay

3.1 Game Flow

The central gameplay loop is split into two parts: Dungeon Mode, and Writing Mode.

In **Dungeon Mode**, the player explores an overworld full of different characters, obstacles and other interactable objects.

Upon interacting with an object the game switches to **Writing Mode**, and the player must write answers to prompts to get past the object.

Completing one prompt will cause another to appear, looping until all prompts are completed and the object is marked 'complete'.

Upon 'completing' an object, the game switches to **Dungeon Mode**, and the player is free to explore. If the object was an obstacle, that obstacle is now cleared, opening up more areas in the world.

The main focus of the game is the writing sections, and so the player can expect to spend more time in Writing Mode than Dungeon Mode, at 65/35 split.

Game flow – play-by-play

1. **Character creation**: players must fill out character details and customise the look of their hero.

2. **Destination choice**: players can choose which destination they want to go to

3. **Dungeon Crawling begins! Tutorial onboarding.** The player's companion introduces them to walking around the overworld, and interacting with objects.

4. **Move Around**: players use the mouse to move around the overworld, and can interact with objects with space.

5. **Interact with object**: When a player interacts with an object, they move to Writing Mode.

6. **Prompt appears**: The player must complete a writing prompt to 'defeat' an obstacle.

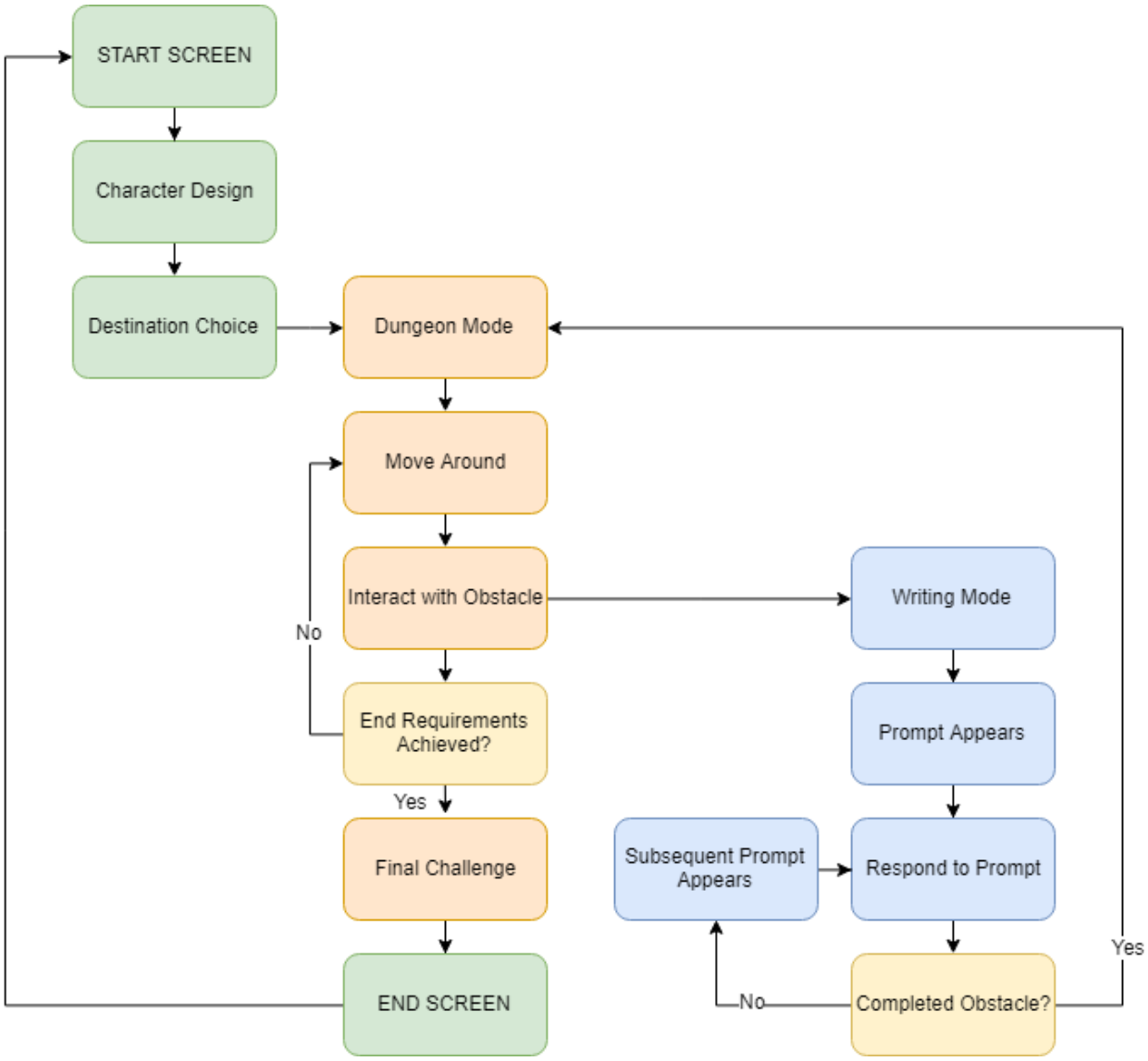
7. The game checks to see if the response to the prompt is adequate to pass the obstacle. If not, a subsequent prompt appears. The player continues to respond to prompts, building up a paragraph response that will fit into their story.

8. Once the obstacle is completed, players return to the **Dungeon Mode** overworld, where they can continue exploring and looping through other obstacles.

9. If all the obstacles have been interacted with, then the players complete a **Final Challenge** (in writing mode). This ends the story

10. **End Screen**: the player is faced with the end screen, on which they can read their completed story (compiled from their writing prompts) and choose to print it before returning to the start screen.

Game flow - diagram



3.2 Objectives

Chapter 1 has both extrinsic and intrinsic objectives, but these objectives are balanced to control the motivation of the player towards the game.

1. Within the game, the objective of the player is to reach one of the exits to the dungeon by passing any obstacles that block their path. This is their extrinsic motivation; to reach the goal and 'beat the game'

2. Outside of the game, the player's objective is to write a story as guided by the prompts that they encounter during gameplay. This can be an intrinsic motivator, if the player enjoys writing. However in many school-age children this is not the case. Therefore, the first objective becomes more important because it a) is more engaging to the average player and b) it enables the second goal.

As a result, the player is not told that they must write a story (objective 2); instead they are told they must reach the dungeon exit (objective 1) – and in order to do that they must write a story.

3.3 Gameplay: Dungeon/Adventure Mode

The Dungeon Crawling Mode (I'm thinking of labelling this 'Adventure Mode' to be more appropriate for young people) is the game's 'overworld'. Players explore the overworld and the things they encounter form the building blocks for their story. Players not only get to write the story, they get to live the story. Each overworld should contain different objects and paths to make the world feel new each time.

Basic Elements

Map: The overworld will have an isometric tile-based map. The tile-based system means it is easy to design levels on a grid, blocking out different pathways for the players to go, and different obstacles they might face on a tile-by-tile basis. The bounds of the map will be marked by walls. The map should not be too big so as to lose a player's attention, but not too small to prevent the creation of a full story.

Playable Character: The player will design their character at the beginning of the game, and this will then be translated into a sprite that they can guide around the overworld.

Obstacles: The player will encounter a variety of obstacles in the overworld that they will have to pass. Interacting with an obstacle will trigger the game to switch to writing mode, upon which the player is prompted to write how they will get past the obstacle.

NPCs: Some worlds may have NPC characters that the player can talk to. These would also prompt the player to write interactions, with prompts geared at conversation.

Inventory Items: I am undecided on including inventory items that can be references by the player in later written prompts. This might be a nice feature, but might complicate things; I want the player to use their writing to solve problems, rather than having an inventory item that does the same thing.

Dialogue Box: The bottom 25% of the screen will be taken up by a dialogue section, which will show dialogue whenever the player moves close to an object, giving added character to the world. The dialogue box will feature an animated avatar of the player character when they speak.

Scoring: I will not be using a score to incentivise playing the game; it is difficult to quantify the player's writing, and the incentive should be the desire to finish your story.

Controls

The overworld controls are intentionally simple, so as not to introduce a steep learning curve – the learning should be focuses on the writing, not working out the gameplay.

Movement: The playable character will move around the map by clicking the tile you wish to move to.

Interaction: When next to an interactable object (indicated by a shine on the object sprite), pressing the space bar will switch to the writing mode.

UI

The UI will be made up of two 'blocks': a window containing the game world and a window containing player dialogue, use to indicate when they are near an interactable object.

The wireframe below indicates the size and layout of these blocks. Colours not indicative of the final UI colours.

Example wireframe

3.4 Gameplay: Writing Mode

The Writing Mode is the core of the game. Upon interacting with an object in the overworld, the player will be prompted to write a response to the object. For example, the object might be a ravine, and the player might have to describe how they get over the ravine. Upon completing a prompt, the player receives additional prompts to follow based on what they first wrote. They might be asked to describe the rope they used to swing across.

After completing a nominal number of prompts, the object will be classed as 'completed' and players will be able to return to the overworld mode to continue their exploration. The completed passage of writing will be added to their story.

Basic Elements

Prompts: Each object will have an initial prompt attached to it, which the player must answer. New prompts are then generated, which relate to what the player has already written.

Writing Window: This is the space in which the player's writing appears as they type. It will function like an ordinary text box, but be style to fit the game's aesthetic rather than a boring Word Doc. Players will be able to submit what they've written with a button click.

Support Prompts: After answering the initial prompts the player may receive addition ideas e.g. 'Try using a new adjective' to support their writing. Not yet clear how 'hands-on' this feature will be.

Imagery: The Writing Mode will also feature a section for showing an image of the object being interacted with, or the player avatar where appropriate.

Natural Language Processing

The writing mode will use natural language processing and parts-of-speech tagging to parse what the player writes and generate new prompts based on what has already been written. This creates the idea of a 'teacher within the game', providing help and support as the player goes.

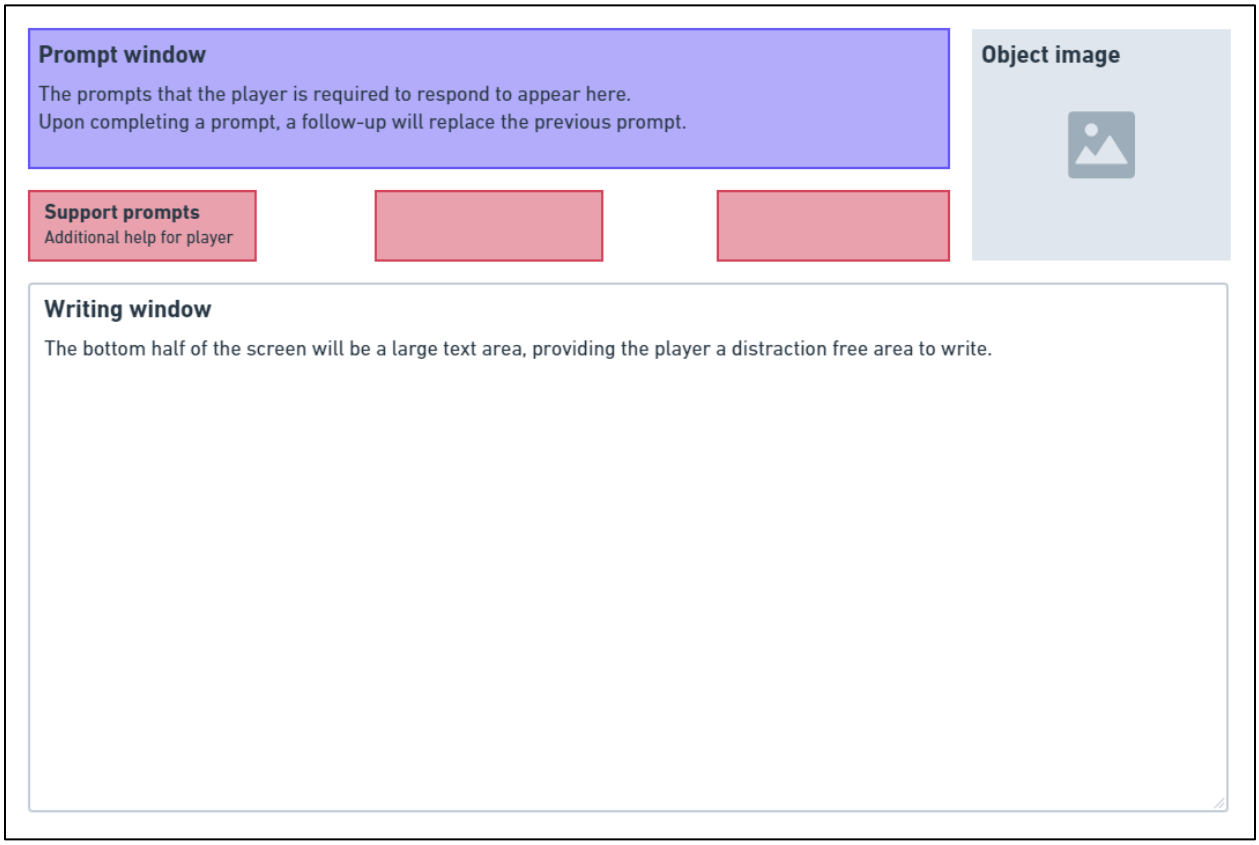
This project will use the RiTa NLP framework for parts-of-speech tagging, and will require an amended dictionary to ensure that tagging is correct.

Natural language processing can also be used to detect the parts of speech that players are using most, and encourage the use of modifiers such as adjectives and adverbs to enhanced their work.

UI

The Writing Mode UI will reverse that of the Dungeon Mode, with a large block at the bottom of the screen containing space for the player to write, with prompts appearing above alongside an image of the obstacle/object the player is writing about.

The wireframe below indicates the size and layout of this mode. Colours not indicative of the final UI colours.

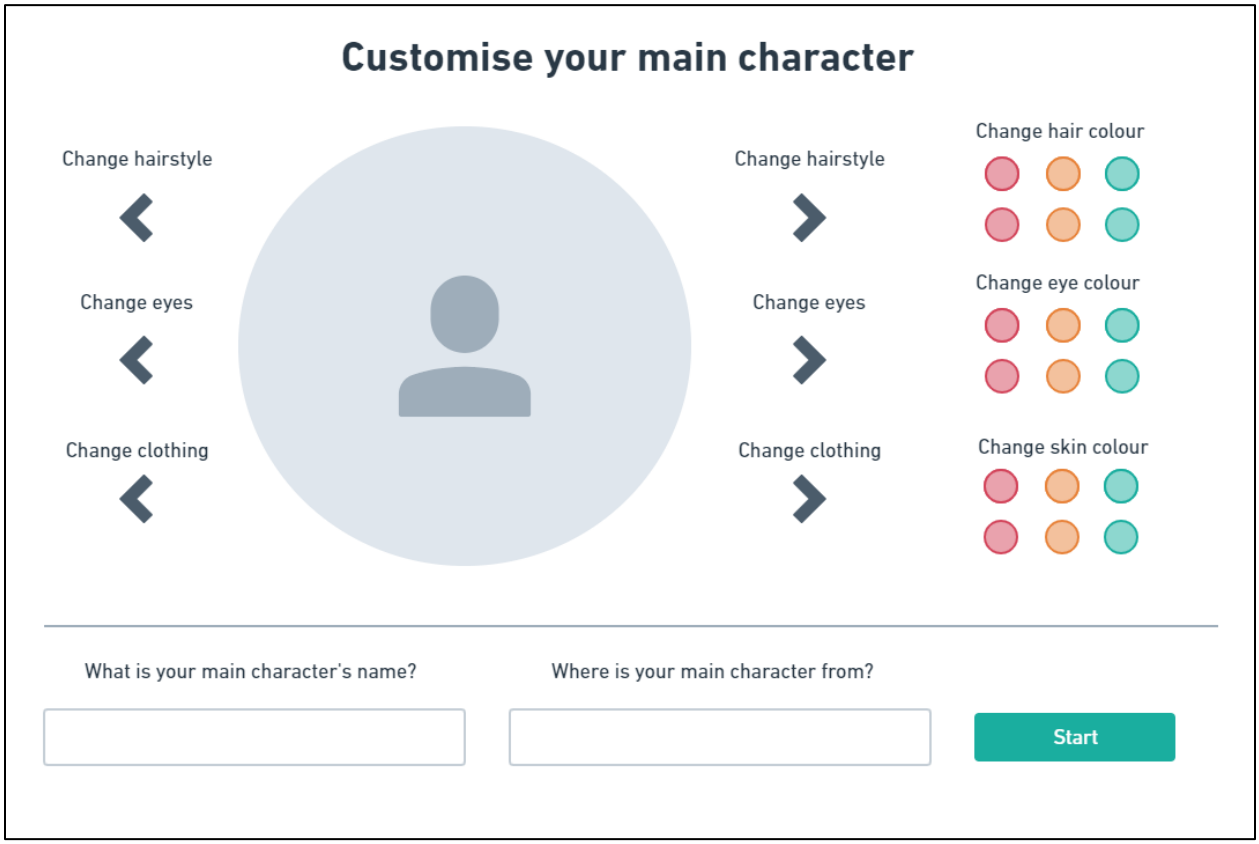


3.5 Character Customisation

Players of the game should feel ownership of the character they are playing in the game – it should feel more than an avatar, but an extension of their story or an identity that they can project themselves onto should they wish. The design workshops showed that by anchoring a part of themselves in the character players will take on more ownership of the story, and it will feel more unique to them.

With this in mind, a character customisation menu is an important part of the gameplay experience. Players will be able to:

- Name the character
- Name where the character is from

- Choose the character's hair style, eyes and clothing
- Change the character's skin tone, eye colour and hair colour.

All players should have enough customisation options to choose from to design the protagonist feel best represents them or their story's protagonist.